

EduGenie – Project Development Phase

5. Project Development Phase

This document describes the development stage of EduGenie and the main application components used to build the learning assistant.

1. Development Technology Stack

Technology	Purpose
Python	Core programming language.
FastAPI	Web application/backend framework used by EduGenie.
Google Gemini API	AI service used to provide AI-generated responses.
Jinja2	Template rendering for the web interface.
HTML, CSS, JavaScript	Frontend interface and client-side behavior.

2. Main Application Features

EduGenie is designed as an AI-powered learning assistant. The project includes educational question answering, simple explanations, study-material summarization, quiz generation, and personalized learning paths.

3. Development Flow

The development flow consists of preparing the web interface, connecting the application to the backend, integrating the Gemini API, implementing the learning-assistant functionality, and organizing the supporting project files.

4. Project Development Files

The repository contains application-related folders such as app, static, templates, and tests, together with configuration and project documentation files.

5. Security Note

API keys and other secrets should not be placed directly in source files or committed to GitHub. The project includes an .env.example file for environment-variable guidance.

6. Development Deliverable

The expected deliverable of this phase is the working EduGenie application together with its source code, templates, static resources, tests, and required configuration files.